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Mohid Abdul Rehman

AI/ML Engineer. Generative AI and LLM project portfolio

I build production AI systems, mostly LLM applications: multi-agent RAG, embeddings and vector search, vision-language extraction, and self-hosted open-source models. Over the past two years I have shipped these at Genetech Solutions and Softstream.ai, from corpus preparation and retrieval design through to APIs and deployment.

About the code

The systems in this document were built for employers and clients and are covered by confidentiality agreements, so their source code is not public. They are described here at architecture level only. I am glad to walk through the design, trade-offs and results in a technical interview. My public code is listed at the end.

Skills in practice

RAG pipelines	Knowledge Hub (multi-agent, citation-backed), TutorBot (document Q&A)
Agentic and multi-agent AI	Knowledge Hub: agent-routed retrieval over Qur'an, tafsir, hadith and fiqh sources
Embeddings and vector search	Smart Job Onboarding, AI Image Search Engine (ChromaDB), Portfolio Smart Search
Self-hosted LLM serving	Qwen served on GPU with vLLM, giving zero per-query API cost
Prompt engineering and VLMs	Dynamic Form OCR: schema-constrained prompting for structured extraction
APIs and deployment	FastAPI (Portfolio Smart Search), Flask (TutorBot), AWS EC2 (AI Exercise Trainer)
LLM APIs and NLP	OpenAI API (TutorBot, Customer Feedback Analysis), speaker diarization, sentiment analysis

GenAI and LLM systems

Knowledge Hub

Multi-agent RAG chatbot

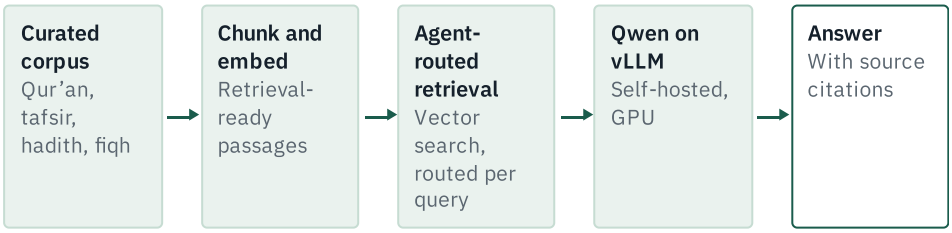
Genetech Solutions
Full-time, 2025 to present

Code under NDA

Citation-backed answers from authenticated sources only

Zero per-query API cost with a self-hosted model

A chatbot that answers Islamic questions using only authenticated sources, namely the Qur'an, tafsir, hadith and fiqh texts across multiple madhhabs, and shows where each answer comes from. I architected it end to end. Instead of a paid API it runs on a self-hosted Qwen model served on GPU through vLLM.



- Owned corpus curation, chunking, embedding and agent-routed retrieval.
- Set up local LLM serving with vLLM so the system has no external API dependency.

Stack Python, Qwen, vLLM, embeddings, vector database, multi-agent orchestration

AI-Powered Smart Job Onboarding System

Semantic candidate–job matching

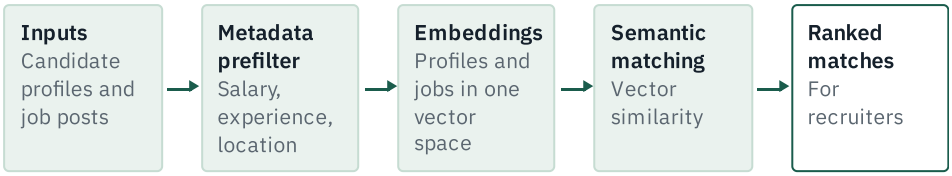
Genetech Solutions
Full-time, 2025 to present

Code under NDA

~80% better candidate–job match accuracy

70% less recruiter screening time

Matches candidates to jobs by meaning rather than keyword overlap, while still respecting hard requirements such as salary, experience and location. I designed and deployed it.



Stack Python, embeddings, vector search with metadata filtering

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Dynamic Form OCR Pipeline

Vision-language extraction from handwritten forms

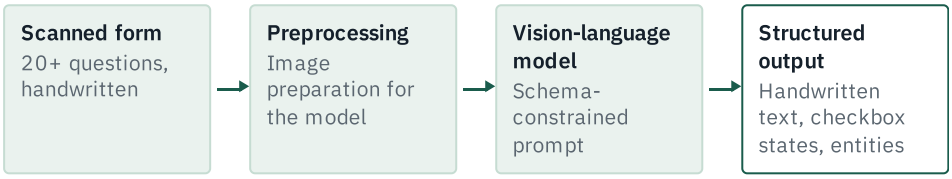
Genetech Solutions
Full-time, 2025 to present

Code under NDA

Manual data entry eliminated

~75% faster form processing

Registration forms with 20+ questions, handwriting and checkboxes were being typed in by hand. This pipeline reads them with a vision-language model and returns structured data, using a schema-constrained prompt so every form comes back in the same shape.



Stack Python, vision-language model, prompt engineering, image preprocessing

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AI Image Search Engine

Multimodal search over 25K+ photos

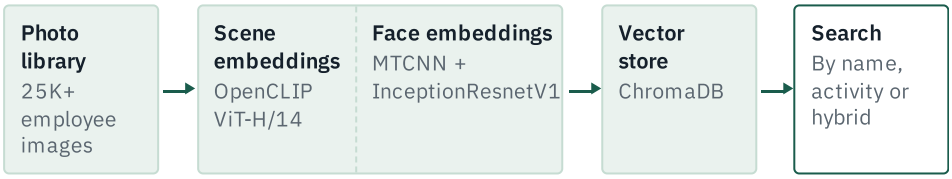
Genetech Solutions
Full-time, 2025 to present

Code under NDA

~85% less manual photo retrieval time

Used by the content and marketing teams

Lets the content and marketing teams find photos in a library of 25,000+ employee images by who is in them, what is happening in them, or both. Each photo gets two kinds of embedding, one for the scene and one for each face, stored together in a vector database.



Stack Python, OpenCLIP, facenet (MTCNN, InceptionResnetV1), ChromaDB

Portfolio Smart Search Engine

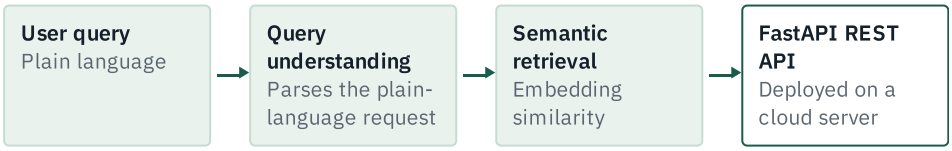
Natural-language search API

Genetech Solutions
Full-time, 2025 to present

Code under NDA

~70% more relevant results than keyword search

Replaces keyword search with search that understands plain-language queries. It is served as a REST API with FastAPI on a cloud server.



Stack Python, FastAPI, embeddings, semantic retrieval, cloud deployment

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TutorBot

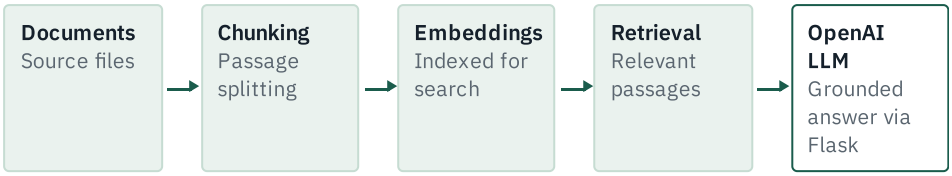
Document Q&A with RAG

Softstream.ai
Internship, 2024

Code under NDA

Full RAG flow built from chunking to answer generation

A chatbot that answers questions from a document collection, grounding each answer in retrieved passages. I built the retrieval pipeline, a Flask backend and a custom frontend.



Stack Python, OpenAI API, Flask, embeddings, custom frontend

Customer Feedback Analysis Tool

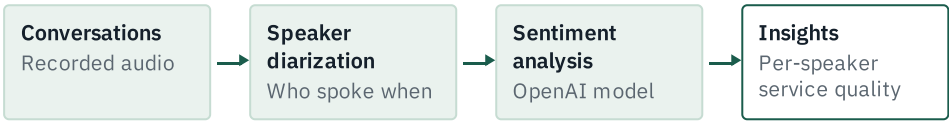
Per-speaker sentiment from conversations

Genetech Solutions
Internship, 2025 (client project)

Code under NDA

Per-speaker view of service quality for the client

Analyses customer conversations by first separating who is speaking, then scoring sentiment for each speaker with an OpenAI model, giving the client a per-speaker view of service quality.



Stack Python, speaker diarization, OpenAI API

Computer vision

Motorbike Theft Detection System

Genetech Solutions, internship, 2025. Code under NDA.

Real-time detection with YOLOv11, multi-object tracking across frames with DeepSORT, and CLIP embeddings for person re-identification and visual similarity matching.

AI-Powered Exercise Trainer

Softstream.ai, internship, 2024. Code under NDA.

MediaPipe body-landmark tracking to evaluate posture and correct exercise form, deployed on AWS EC2.

Public code on GitHub

facial-recognition

Real-time webcam face recognition in modular Python: YOLOv11 face detection, GFPGAN face enhancement, and InceptionResnetV1 (VGGFace2) embeddings matched by cosine similarity.

EARLY-THEFT-DETECTION

Shoplifting detection on surveillance video using an LRCN (CNN + LSTM) sequence model, writing an annotated output video with a risk message.

Machine-Learning-Projects-Collection

Prediction, classification and clustering projects with preprocessing, exploratory analysis and Plotly/Seaborn visuals.

Pose-Master

Flask web app that tracks MediaPipe pose landmarks, computes joint angles to check exercise form, and gives spoken feedback.

Deep-Learning-Projects-Collection

ANN, CNN and RNN notebooks, including plant disease diagnosis, VGG16 transfer learning and sentiment analysis.

Kaggle profile

Notebooks published at kaggle.com/mohidabdulrehman.

Education

BS Computer Science, Federal Urdu University of Arts, Science & Technology, Karachi (2021 to 2025, CGPA 3.5/4.0)

NAVTTC AI (ML/DL) course, DHA Suffa University